



High Availability of OpenEMR in Cloud (AWS)



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1. Introduction

In spite of the accelerating migration to cloud computing in the Healthcare Industry, high availability and uptime concerns rank as one of the top challenges for Healthcare Organizations looking to adopt cloud services, states the 2014 HIMSS Analytics Cloud Survey report.

This document focuses on achieving High Availability for OpenEMR in Cloud (AWS). It focuses on how a stable environment for an EMR (OpenEMR) could be offered in a cloud environment (AWS). The importance of cloud for healthcare IT and High Availability solutions for OpenEMR application in cloud is explained.

OpenEMR is one of the most popular Open Source EMR for practices that has been deployed significantly worldwide.

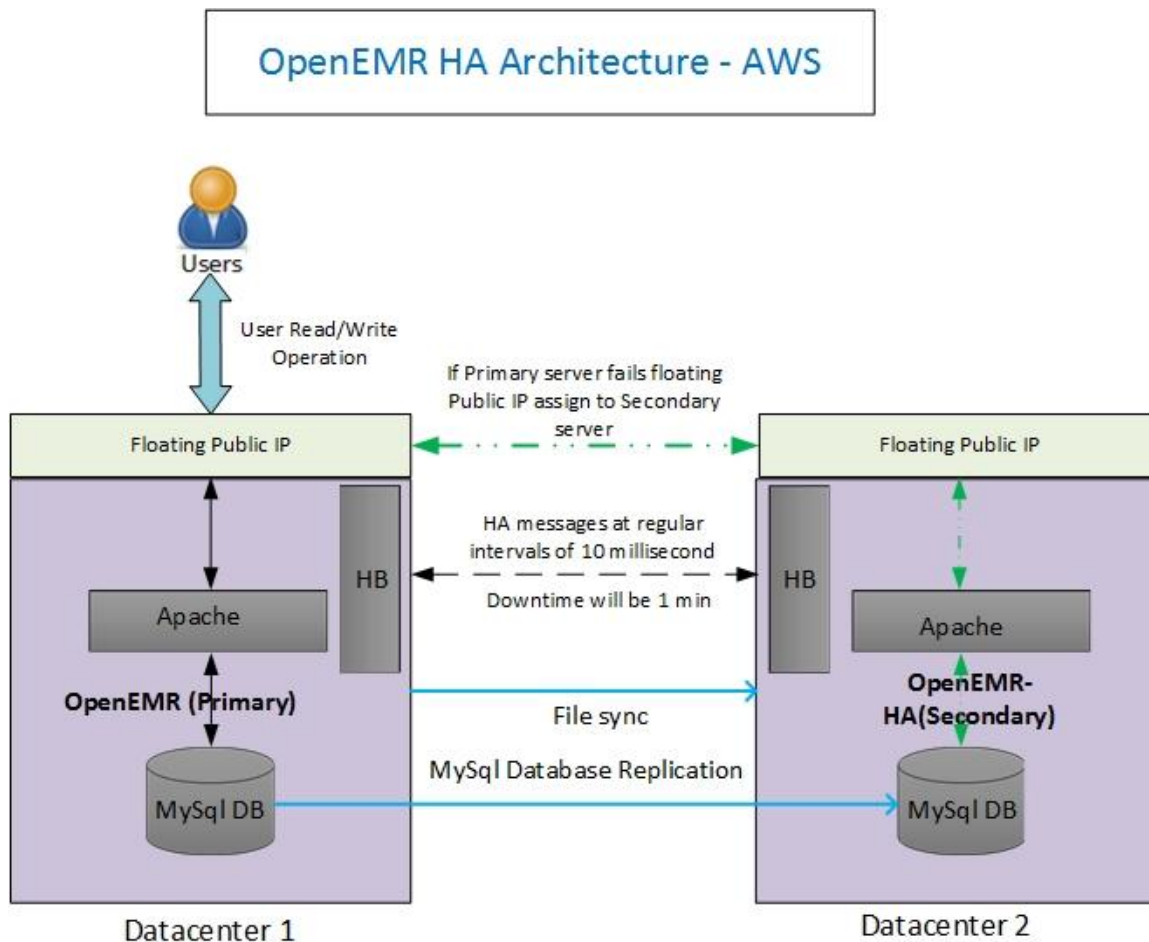
Please refer to the Technical Guide titled “Hardening OpenEMR in Cloud (AWS)” on details regarding how to create a new AWS instance and setup OpenEMR within the cloud environment.

2. Need for High Availability of OpenEMR in Cloud

Hosting OpenEMR in Cloud (AWS) can be a cost efficient decision with several benefits. Maintaining the data locally, increases the possibility of disk data loss and theft which is a security breach as per the HIPAA standards. Most of the legacy or existing infrastructure that is designed to handle limited or specific number of users may not support the performance of applications when the staff or users in healthcare practice increase in size. Hardware failure can occur any time. Multiple hardware failure like CPU failure, memory failure and disk corruption requires hardware to be replaced and infrastructure to be upgraded after a specific period of time so that the performance and service standards are not compromised. So hosting OpenEMR in Cloud is definitely a better choice that can enable better business outcomes.

High Availability of OpenEMR in Cloud is even more crucial because physicians need to have access to OpenEMR application round the clock to be able to take timely decisions and provide enhanced patient care. So ensuring High Availability of OpenEMR in Cloud through Open Source Technologies as emphasized in this document, enables physicians not just to provide timely care but also empowers to save lives during mission critical situations and medical emergencies.

2.1 HA Architecture & Component Overview



2.1.1 Heartbeat

- Download and install Heartbeat software in the primary server and secondary server.
- Configure heartbeat to monitor the following services
 - floatingIP – Attach/Detach the floating IP (Global IP) for clients to connect to the server
 - httpd /Apache– Web server
 - nginx – High performance HTTP and reverse proxy server
 - MySQL DB- Database that syncs the data between both servers
- Heartbeat runs as master in one server and slave in another server. In master server, all the services will be up & running (httpd/nginx/cron/etc). Whereas in slave server, the services will be in stopped state apart from MySQL DB.

2.1.2 MySQL Replication

- Configured MySQL database replication from primary server to secondary server and vice versa.
- When primary server is active (master), clients will be connecting to primary server and data will be updated accordingly. When any insert/delete/update operation is performed in primary server, it will be immediately (10 microseconds) synced/replicated to secondary server.
- Similarly, when secondary server is active, clients will be connecting to secondary server and data will be updated in secondary server database. When any insert/delete/update operation is performed in secondary server, the same will be synced/replicated to primary server.
- MySQL needs to be up and running in both the servers for replication. So it will NOT be configured in Heartbeat for monitoring.

2.1.3 Synchronization

- Configure web files/documents/folders synchronization from primary server to secondary server and vice versa.
- When primary server is active (master), synchronization will happen from primary server to secondary server.
- When secondary server is active (master), synchronization will happen from secondary server to primary server.

2.1.4 Workflow representation

The architecture in high availability holds two machines, primary and secondary machine or master and slave connected to network. The primary machine stands as a master server that processes continuous requests, updates and other process requests from the EMR application.

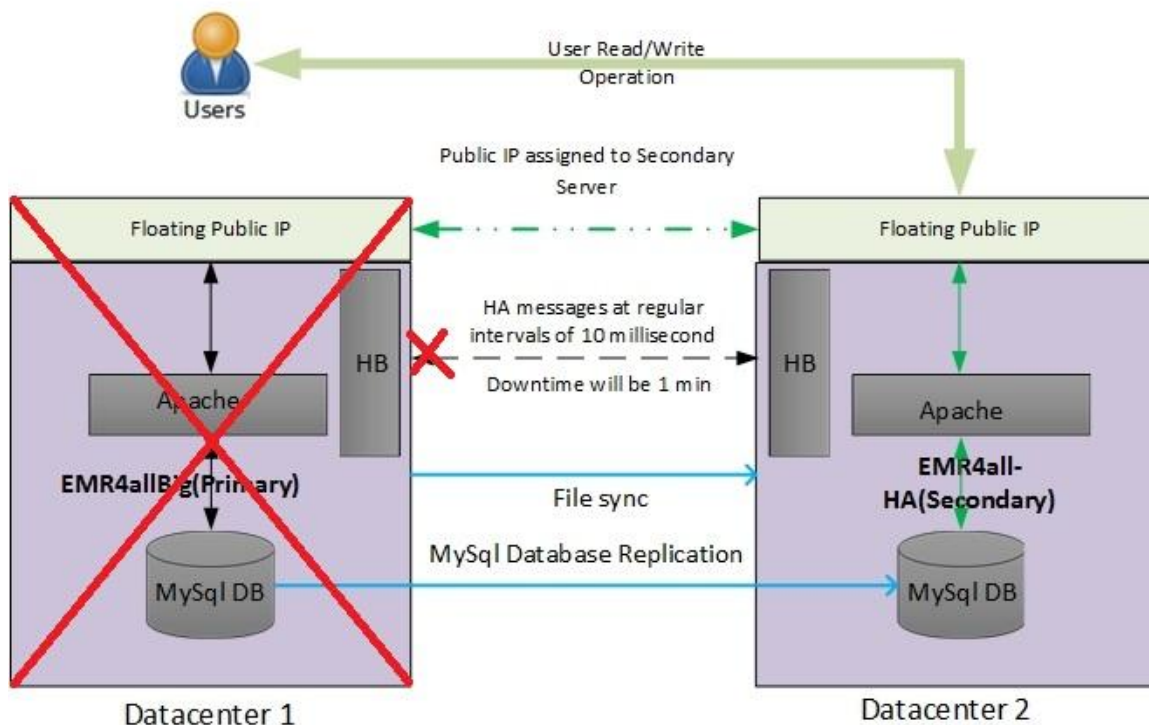
The requests and response from the EMR to master server is continuously updated and saved at master server. The secondary server that is connected with the primary server, pings or checks for the existence of the primary machine with the programmed interval. This ensures that the primary machine is up and process the requests with the application without any concerns.

As per the scheduled interval, the data present in the primary server is replicated or updated in the secondary server to act as a backup machine during downtime. When the primary server is down, the secondary server acts as a primary server and process the applications data. This ensures minimal data loss and downtime.

2.2 Handling Server Failure:

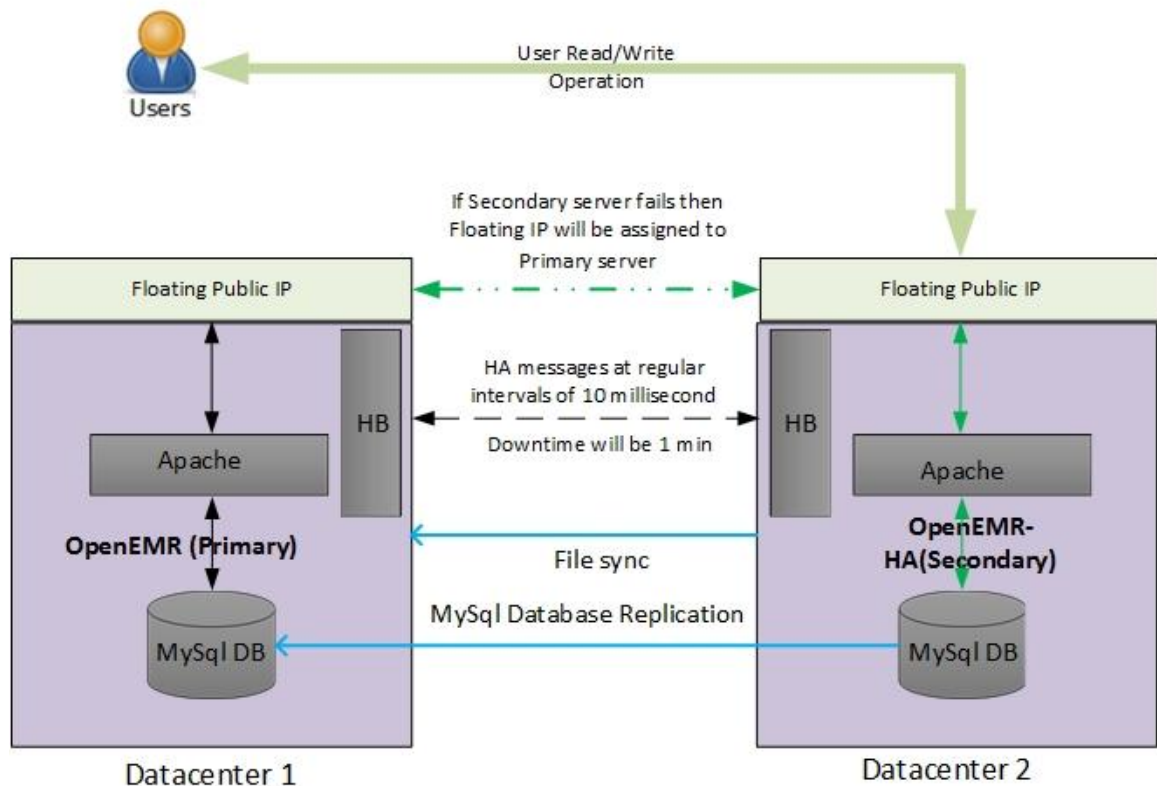
The failure of the main server is recognized by continuous heart beat signal that is processed between the primary server and the secondary server. When the main server fails, the secondary server picks up the IP of the main server that has failed and acts as a primary server. With this implementation, the downtime of the system or the unavailability of the system is reduced. At this point, the secondary system acts as a primary system and process all data requests and response from the EMR application. This is continued till the primary server is up and ready for data transactions from the EMR application.

Server Failure



2.3 Restoring the server after failure

Server Restore



Once the initial server that was dropped is restored, the secondary server which acts as a primary server remains to act as the same. The restored server continuously checks the existence of other server using heart beat signal that is processed between both the systems and acts as secondary server that replicates the updates happening with the currently active server.

3. Conclusion

High Availability of EHR Applications can be affected due to hardware and software failures. Firstly, by moving the infrastructure to the Cloud it is assured that the cloud service will be responsible for hardware availability. In terms of software availability for OpenEMR, implementing High Availability Solution as described in this documents ensure that the physicians can have 7x24 access to OpenEMR to take timely medical decisions and to improve overall patientcare.



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